

# Vasileios Evangelopoulos

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 GitHub |  Website

## EDUCATION

### University of Patras

October 2020 – October 2025

*MEng in Electrical and Computer Engineering*

*Patras, Greece*

- GPA: 8.22/10.0
- 5 year program with integrated Master's degree
- **Thesis:** *Control design of HVDC link with virtual synchronous generator and implementation on real-time simulator*

## EXPERIENCE

### Undergraduate Research Assistant

June 2020 – Present

*Texas A&M University*

*College Station, TX*

- Developed a REST API using FastAPI and PostgreSQL to store data from learning management systems
- Developed a full-stack web application using Flask, React, PostgreSQL and Docker to analyze GitHub data
- Explored ways to visualize GitHub collaboration in a classroom setting

### Information Technology Support Specialist

Sep. 2018 – Present

*Southwestern University*

*Georgetown, TX*

- Communicate with managers to set up campus computers used on campus
- Assess and troubleshoot computer problems brought by students, faculty and staff
- Maintain upkeep of computers, classroom equipment, and 200 printers across campus

### Artificial Intelligence Research Assistant

May 2019 – July 2019

*Southwestern University*

*Georgetown, TX*

- Explored methods to generate video game dungeons based off of *The Legend of Zelda*
- Developed a game in Java to test the generated dungeons
- Contributed 50K+ lines of code to an established codebase via Git
- Conducted a human subject study to determine which video game dungeon generation technique is enjoyable
- Wrote an 8-page paper and gave multiple presentations on-campus
- Presented virtually to the World Conference on Computational Intelligence

## PROJECTS

### B-H Curve using Arduino | *Python, Matlab, Arduino, FEMM, Circuit Design*

June 2020 – Present

- Captured the hysteresis loop of a transformer core material using the Arduino UNO microcontroller
- Designed circuits for measuring primary winding current and secondary voltage
- XY plot using Matlab and Manim python library for visualization

### FEA of a Reluctance Machine | *FEMM 4.2, Matlab, AutoCAD, Blender*

May 2018 – May 2020

- Designed a synchronous reluctance machine in AutoCAD
- Implemented a Matlab script to control FEMM software
- Post-processed FEMM data to calculate various performance parameters
- Designed a 3D model for visuals

### Differential Drive Robot | *ROS, Python, Arduino, Linux*

May 2018 – May 2020

- Implemented state estimation for a differential drive robot using Dead Reckoning method
- Developed a localization system by implementing Adaptive Monte Carlo (AMCL) via Particle Filters
- Visualized real-time sensors data and system states using RViz
- Programmed firmware using Arduino

### Monitoring & Control Propane Levels in a Tunnel | *LabVIEW, Arduino*

May 2018 – May 2020

- Implemented monitoring and control system using LabVIEW
- Developed a gas actuation mechanism and a tunnel apparatus

## PUBLICATIONS

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- V. Evangelopoulos**, M.-E. Karaitsi, T. Kavvathas, T. Alexandridis, G. C. Konstantopoulos 2025
- *"Combined Synchronverter-VSM based HVDC transmission design with Power-Hardware-In-the-Loop (PHIL) implementation"*, 2025 IEEE PES Conference on Innovative Smart Grid Technologies - Middle East (ISGT Middle East), Dubai, United Arab Emirates, doi: 10.1109/ISGTMiddleEast65737.2025.11314231.
- N. Bali, **V. Evangelopoulos**, M.-E. Karaitsi, P. Bouras and J. Gialelis 2025
- *"Human-centric approach for determination of Thermal Plume theoretically and experimentally"*, 2025 IEEE 30th International Conference on Emerging Technologies and Factory Automation (ETFA), Porto, Portugal, doi: 10.1109/ISGTMiddleEast65737.2025.11314231.

## TECHNICAL SKILLS

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**Programming Languages:** Matlab/Simulink, Python, LabVIEW, C++, HTML/CSS  
**Software:** FEMM, Typhoon HIL, AutoCAD, Multisim  
**Tools:** Git, GitHub, Linux, L<sup>A</sup>T<sub>E</sub>X, ROS